

Shopping Calculator

A Practical Study in Mixing Java Data Types

1. The Logic Breakdown

In this example, we see how `int`, `float`, and `char` come together to create a functional billing logic.

`int items = 50;`

Whole numbers are used for counting.
You can't buy half an item in this logic.

`float cost = 9.99f;`

Decimals are required for currency.
Don't forget the
`'f'`
!

`char currency = '$';`

A single symbol representing the unit
of measurement.

`float total = ...;`

The result of multiplying an `int` and a
`float` must stay as a decimal.

2. Implementation Source Code

ShoppingCalculator.java

```
public class Main {  
    public static void main(String[] args) {  
  
        int items = 50;  
        float costPerItem = 9.99f;  
        float totalCost = items * costPerItem;  
        char currency = '$';  
    }  
}
```

```
System.out.println("Total cost = " + totalCost + currency);  
}  
}
```

Codehive

3. The Program Output

When the code runs, Java performs the math and joins (concatenates) the numbers with the text and currency symbol.

```
Number of items: 50  
Cost per item: 9.99$  
_____  
Total cost = 499.5$
```

4. Advanced Formatting

By default, Java might show 499.5. In a real store, we usually want 499.50. You can achieve this using `printf`:

```
System.out.printf("Total: %.2f%c", totalCost, currency);
```

(Note: `%.2f` tells Java to show exactly 2 decimal places)

5. Key Summary

Why choose these types?

- **int**: Perfect for counts/quantities.
- **float / double**: Essential for prices and scientific math.
- **char**: Ideal for one-room labels like symbols (+, -, %, \$).
- **Combining**: Java allows different numeric types to interact in the same math equation.